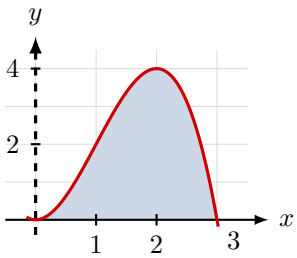
L4.3b Which Method?

Work in your group. Check your answers against the Quarto tonight.

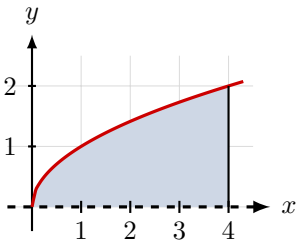
Choosing the method

1. Each shaded region is rotated about the dashed axis. Decide first: which variable are the curves written in? Draw the rectangle, name the method, and write the integral. Do not evaluate.

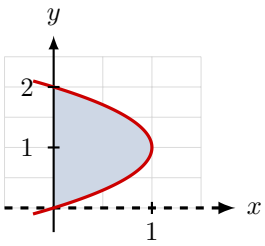
(a) $y = 3x^2 - x^3$, $y = 0$, about the y-axis



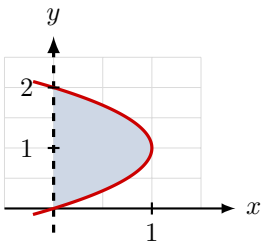
(b) $y = \sqrt{x}$, $0 \leqslant x \leqslant 4$, about the x-axis



(c) $x = 2y - y^2$, $x = 0$, about the x-axis

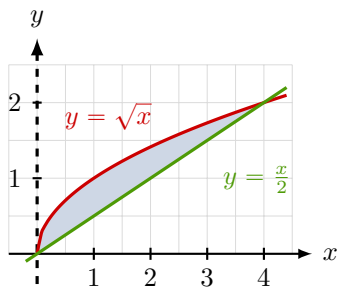


(d) $x = 2y - y^2$, $x = 0$, about the y-axis



One solid, two methods

2. Decide first. The region between $y = \sqrt{x}$ and $y = \frac{x}{2}$ is rotated about the y-axis.



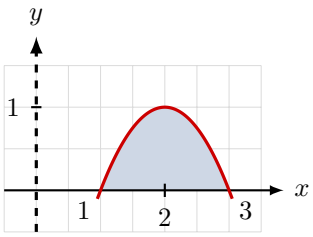
(a) Find the volume with shells.

(b) Find the volume with washers.

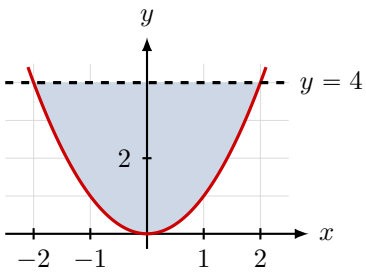
(c) Which was easier? Why?

Volumes by any method

3. Decide first. The region bounded by $y = -x^2 + 4x - 3$ and $y = 0$ is rotated about the y-axis. Find the volume.



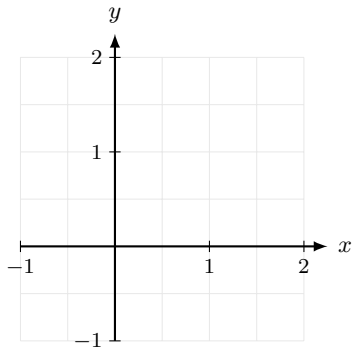
4. Decide first. The region bounded by $y = x^2$ and $y = 4$ is rotated about the line $y = 4$. Find the volume.



5. Each integral is the volume of a solid. Name the method, the region and the axis of rotation, then evaluate the integral.

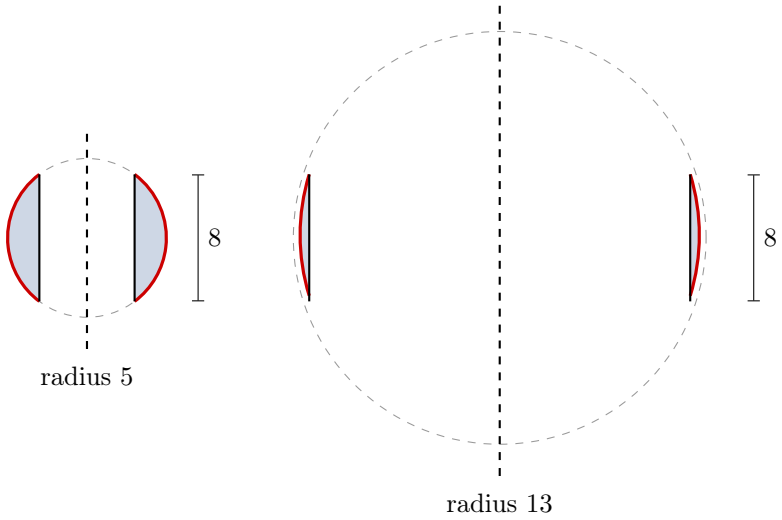
(a) $\pi \int_0^1 x - x^4 \, dx$

(b) $2\pi \int_0^1 x\left(\sqrt{x} - x^2\right) dx$



Napkin rings

6. BOARDS Two wooden balls have radius 5 cm and radius 13 cm. A cylindrical hole is drilled through the center of each, so that both napkin rings are 8 cm tall. Which ring has more wood? Guess first, then find the volume of each ring.



Each picture is a slice through the center: the shaded parts are the wood.

What if the rings are h cm tall instead?